

AI Language Cleanup Report

Date: March 15, 2026
Scope: All 40 markdown files in `/home/ubuntu/ems-optimization`

Approach

- 1. **Automated pass** — A Python script (`cleanup_ai_language.py`) applied 70+ regex/phrase replacements across all 40 `.md` files, targeting common AI-generated patterns.
- 2. **Manual review** — Each file was read and checked for subtler patterns the script couldn't catch (sales-y phrasing, vague qualifiers, formulaic transitions).
- 3. **Final consistency scan** — A full grep sweep confirmed no remaining AI-pattern artifacts.

Patterns Targeted

Category	Examples Replaced
Filler qualifiers	"comprehensive" → "full/detailed/thorough", "robust" (non-technical) → "holds up/reliable"
AI enthusiasm	"exciting", "remarkable", "cutting-edge", "transformative", "game-changing"
Corporate-speak	"leverage" → "use", "harness", "synergy", "empower", "unlock"
Weak transitions	"It is important to note that" → removed, "It should be noted that" → removed
Wordy constructions	"In order to" → "To", "A total of" → removed, "Due to the fact that" → "Because"
Passive filler	"It is worth mentioning" → removed, "plays a crucial role" → "matters"
Sales language	"dominates" → "wins", "transformative" → "sharp", inflated benefit claims toned down

Files Modified

Heavy changes (automated + manual)

- `docs/technical_report.md` — 44 automated + 5 manual fixes (largest file, 1084 lines)
- `docs/executive_presentation.md` — 9 manual fixes (most sales-y language)
- `docs/decisions_log.md` — 6 automated + 1 manual fix
- `README.md` — 5 automated + 1 manual fix

Moderate changes

- `docs/project_workflow_wbs.md` — 3 automated + 4 manual
- `docs/executive_summary.md` — 1 automated + 3 manual
- `docs/project_alignment_verification.md` — 6 manual fixes
- `docs/research_questions_assessment.md` — 2 automated + 2 manual

Light changes (1-2 fixes each)

- `docs/final_summary.md` — 1 automated + 2 manual
- `docs/assumptions_log.md` — 1 automated + 1 manual
- `docs/file_inventory.md` — 2 automated
- `docs/blocker_log.md` — 1 automated
- `docs/cbd_robustness_analysis.md` — 1 automated
- `docs/project_charter.md` — 2 automated
- `docs/queue_analysis.md` — 1 automated
- `docs/gap_closure_report.md` — 2 manual
- `docs/gap_remediation_plan.md` — 3 manual
- `docs/eda_and_data_split_summary.md` — 1 manual
- `docs/implementation_roadmap.md` — 1 manual
- `docs/project_archive.md` — 2 manual
- `docs/output_analysis.md` — 1 manual
- `docs/experimental_design.md` — 1 manual

Not changed (already clean)

- `docs/alternative_analyses_summary.md`
- `docs/capacity_sensitivity_analysis.md`
- `docs/cbd_comparison_and_validity_report.md`
- `docs/cbd_definition.md`
- `docs/cbd_focused_optimization_analysis.md`
- `docs/code_documentation.md`
- `docs/conceptual_model.md`
- `docs/demand_model_spec.md`
- `docs/distance_metric_comparison.md`
- `docs/firehouse_capacity_analysis.md`
- `docs/optimization_formulation.md`
- `docs/optimization_results.md`
- `docs/phase21_assessment.md`
- `docs/phase_comparison_summary.md`

- `docs/service_model_spec.md`
- `docs/source_manifest.md`
- `docs/verification_log.md`
- `results/optimization/PHASE3_SUMMARY.md`

Notes

- **“Robust/robustness”** was kept in statistical contexts (experiment names like “CBD Robustness Analysis”, sensitivity analysis conclusions) since it’s standard technical vocabulary. Only changed when used as a generic filler adjective.
- **File names and experiment names** were never altered.
- **Technical accuracy** was preserved throughout — no statistical claims, numbers, or methodology descriptions were modified.
- The automated cleanup script is at `cleanup_ai_language.py` in the project root (can be safely deleted).